Flight Dynamics Simulation

- Author: Aerospace Eng · MSc
- Aircraft: Cessna 172-Class
- Platform: MATLAB / Simulink
- Method: Small-Perturbation Theory

5.60 rad/s	ζ = 0.580
Short-Period ωn	Damping Ratio

■ PROJECT OVERVIEW

Developed a full **longitudinal flight dynamics simulation** of a Cessna 172-class aircraft in MATLAB/Simulink using small-perturbation theory. Aerodynamic derivatives derived from published wind-tunnel data, validated against NACA/NASA flight test records. Development aligned with **Airbus V-Model simulation standards**.

■ STATE-SPACE FORMULATION

State vector: **x = [u, w, q, θ]■** | Control input: **δe** (elevator)

Derivative	Value	Physical Meaning
Xu	−0.03969 s^{-1}	Axial velocity damping
Zw	−2.5302 s^{-1}	Heave velocity restoring
Mw	−0.35483 $(\mathrm{m \cdot s})^{-1}$	Pitch stiffness
Mq	−3.9652 s^{-1}	Pitch rate damping
Zδe	−12.800 m/s²/rad	Elevator normal force
Mδe	−43.910 rad/s²	Elevator pitching moment

■ SHORT-PERIOD MODE

Parameter	Value	Requirement
ωn (nat. freq.)	5.60 rad/s	—
ζ (damping)	0.580	0.35–1.30 ✓
Period T_SP	1.38 s	—
CAP	PASS ✓	MIL-SPEC met

■ SIMULATION INPUT PROFILES

Step	Steady-state pitch response & trim sensitivity
Impulse	Free-damped short-period oscillation → ζ measurement
Doublet	Flight-test standard; excites SP mode cleanly
Sinusoidal	Frequency-sweep → resonance mapping

■ EIGENVALUE STABILITY ANALYSIS

Full 4th-order eigenvalue decomposition — **short-period** (rapid pitch) and **phugoid subsidence** (slow speed-altitude exchange) modes identified.

Mode	Eigenvalue λ	ωn	ζ	Status
SP 1	`−3.248+4.558j`	5.597	0.580	■ STABLE
SP 2	`−3.248−4.558j`	5.597	0.580	■ STABLE
Subs.	`−0.0397+0j`	0.0397	1.000	■ STABLE
Alt. Mode	`0.000+0j`	—	—	■ NEUTRAL

All real parts ≤ 0 — Lyapunov stability criterion satisfied.

■ VALIDATION & VERIFICATION (6/6 PASS)

#	Criterion	Result
#1	**Aerodynamic Derivatives**	Validated vs. NACA/NASA wind-tunnel data
#2	**Short-Period ωn**	Analytical 5.60 = Eigenvalue 5.597 rad/s ✓
#3	**Damping Ratio ζ**	0.580 ∈ MIL-SPEC [0.35, 1.30] ✓
#4	**Eigenvalue Signs**	All Re(λ) ≤ 0 — Lyapunov stable ✓
#5	**Phase Plane**	Doublet → converges to equilibrium ✓
#6	**Time-Domain Response**	Matches classical underdamped 2nd-order ✓

■ SKILLS DEMONSTRATED

Skill / Area	Demonstrated Capability
Flight Dynamics	State-space 4-DOF longitudinal EOM
MATLAB/Simulink	Modular subsystems, aerodynamic blocks, ODE integration
Stability Analysis	Eigenvalue decomp., phase-plane, MIL-SPEC verification
Aero Validation	Cross-ref. NACA/NASA & classical textbook criteria
Eng. Standards	V-Model aligned with Airbus simulation standards
Tech Communication	Interactive report: LaTeX, live plots, data tables

[1] Etkin & Reid, Dynamics of Flight (1996) · [2] Stengel, Flight Dynamics (2004) · [3] Cook, Flight Dynamics Principles (2013) · [4] NACA/NASA Cessna 172 Data · [5] MIL-HDBK-1797

#FlightDynamics #MATLAB #Simulink #AerospaceEngineering #Hiring